

# Individualised Strength

How MicroPulse turns your chosen exercises, each player's data, and the match countdown into one daily session — and how you keep control of every part of it.

For coaches + S&C

Coach → Strength

~20-min micro-dose

01 – The one idea

## One session a day. You choose how it's built.

Every player gets a single strength session. You pick whether it's the standard team template you set up yourself, or an individualised version that adds each player's own layer — built from the exercises *you* chose for the team.

MODE A

### Standard

The matchday template (MD-4 heavy, MD-3 power ...), tuned to the day and the player's readiness. The classic session — you enter it.

MODE B · TEAM DEFAULT

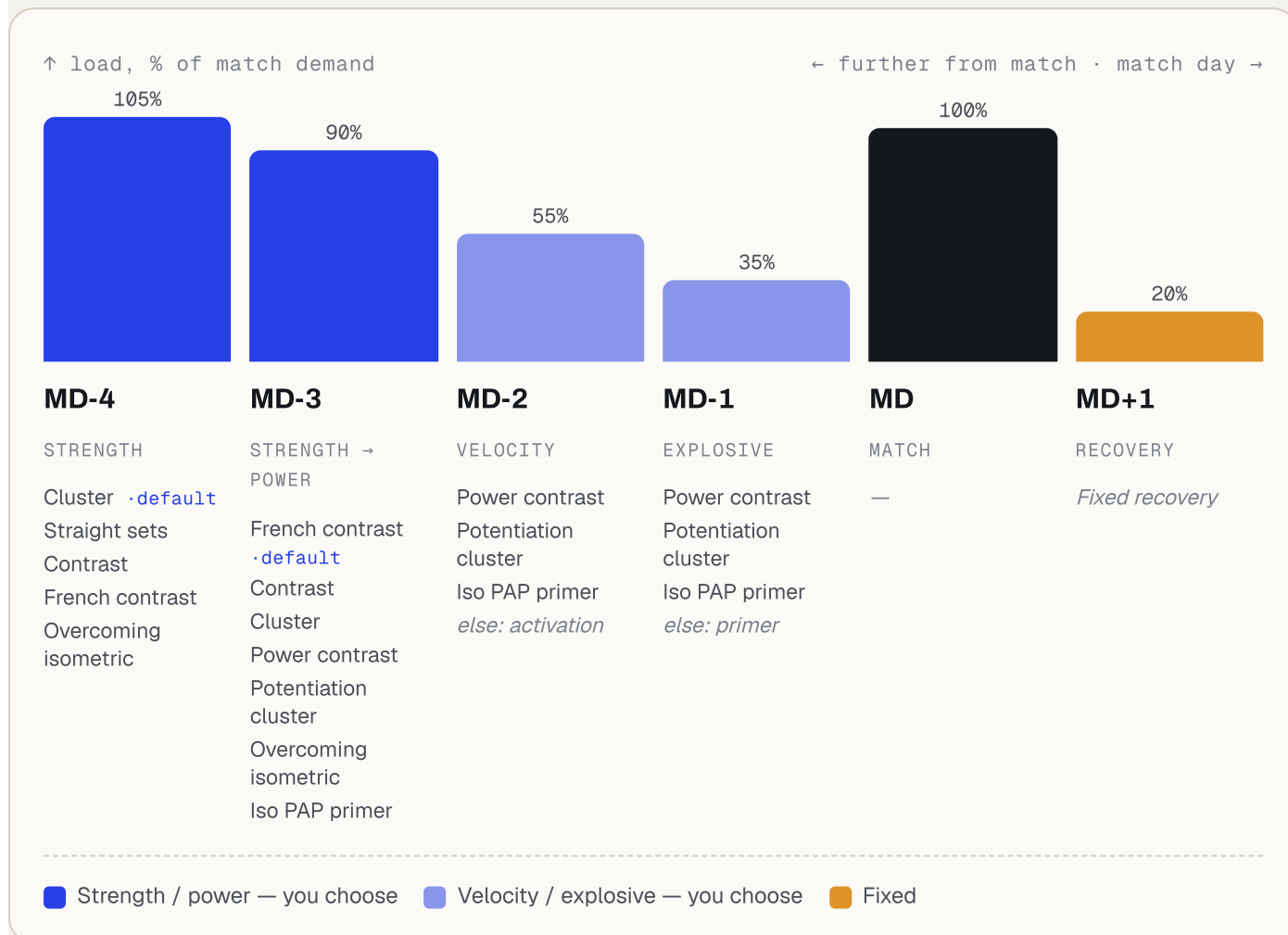
### Individualised

The same template *plus* the player's own layer: their movement-screen corrective, their strength/power deficits, and their left/right symmetry — from your palette.

**Why one session?** In-season, 15–20 minute micro-doses 3–5×/week preserve strength and power without next-day fatigue. [Rønnestad 2023](#).

## Pick the method per day. The load already follows the countdown.

You tie a training *method* to each day; MicroPulse lays it out from your palette. The days near the match only offer light, fast methods — the intent there is speed, not load. Heavy methods are locked to MD-4 / MD-3 on purpose.



Bars show each day's load as % of match demand — the taper the periodisation already applies. [Martin-García 2018](#). Leave a day on *Default* to keep its built-in structure; injury-prevention work (Nordic, Copenhagen) always stays on the strength days.

**One method per day — and it adapts.** You don't stack methods (that bloats the micro-dose). Instead the chosen method **steps down a rung for a yellow player** — French contrast → Contrast, Contrast → Cluster. Green keeps it, red is already recovery, and the set-reduction still applies on top. Two team-mates, same day: the recovered one gets the complex, the tired one a lighter version.

### 03 — Your palette

## Pick the pool. The system picks per player.

The individualised (and auto) build assembles each player's work from a pool you choose. Pick **both** a bilateral and a unilateral lower-body option — the system chooses between them per player from their left/right symmetry.

#### Power / explosive

Jumps · olympic · ballistic · med-ball

#### Bilateral strength

Squat · trap-bar · hip thrust

#### Unilateral strength

Split squat · B-stance RDL

#### Isometric · strength / RFD / PAP

IMTP · iso squat 90° · Spanish squat

**Symmetry decides the lower-body lift.** A player  $\geq 10\%$  left/right asymmetry gets the single-leg lift to load the weaker side; a symmetric player gets the bilateral lift. [Bishop 2020](#). Leave a slot empty → the built-in default. The palette never touches the Standard session.

**Isometrics — for strength *and* power.** Two methods you can tie to a day: **Overcoming isometric** (strength days) — push maximally against an immovable bar at a deep joint angle for max strength + rate of force, joint-friendly; and the **Isometric PAP primer** (taper days) — a maximal isometric conditioning set that potentiates the explosive work 3–6 min later. [Oranchuk 2023](#) · [Krzysztofik 2023](#) · [Jarosz 2025](#).

## Swap, but only for a safe alternative.

When the sent exercise isn't quite right — equipment, comfort, a niggle — both sides can swap it, always from that exercise's curated safe alternatives.

### YOU · THE COACH

On a player's session, hit  **swap**. Safe alternatives are marked and listed first, then the rest of the same category.

### THE PLAYER

A “**Swap an exercise (safe options)**” panel on their Today card — safe alternatives only, never anything inappropriate. Every player swap is logged for you.

## Let it send every morning — if you want.

MicroPulse can send each player their session automatically on strength days, no click. It's self-service per team and  **off by default**

- **You turn it on/off** for your own team on Coach → Strength.
- **Never overwrites** a session you already sent that day.
- **Skips** off-days, injured players, and anyone on a RECOVERY verdict.
- You can always re-send, override, or switch it off.

## Rules decide. The system explains. You have the last word.

- **Every change shows its “why”** and a paper citation under “*Why these changes?*” — a symmetry-driven unilateral lift, a chosen method, a corrective.
- **Your manual swap overrides any system choice** — the engine recommends, you decide.
- **Nothing is hidden:** mode, palette, structures and auto-send are all your explicit settings. The player sees one session, labelled “Sent by your coach”, tuned to how they are today.

## Under the hood

For S&C staff and developers — the pipeline, thresholds, data model and evidence behind the coach-facing behaviour above.

### Build pipeline

`buildStrengthSession(snapshot, coachOverrides, { mode })` — one player, one day:

- 1 **Template / structure** — default → `pickTemplate(md)`; a chosen non-default allowed method → `buildStructuredBlocks(key, md)`.
- 2 **Adaptation rules** — readiness, soreness, decel burden, VBT, sprint drop, CoD asymmetry, congestion. Standard mode drops the deficit-ledger emphases.
- 3 **Team palette** (individualised only) — substitutes your exercises into the power/strength slots at the MD dose; symmetry picks the pool.
- 4 **Coach overrides** — your manual swaps win, last.
- 5 **Correctives** — the movement-screen corrective merges as the front activation block, not readiness-reduced.
- 6 **Serialize** → the Today-card structure, one row per player per day.

## Symmetry & structures

- `codAsymmetryPct ≥ 10%` (Bishop 2020) → unilateral pool, else bilateral. **No cross-pool fallback** — an empty pool leaves the template exercise, so the palette never undoes the engine's own CoD-asymmetry swap.
- Explosive taper methods force a light + fast dose (10% velocity cap) regardless of the day and **skip** the Nordic/Copenhagen block; strength methods keep it. Prevention is never palette-driven.
- Dose is exercise+MD specific and sparse, so a method landing on an undosed day resolves to the nearest MD dose.
- **Readiness downgrade** — a **MODIFIED** / **REDUCED** verdict steps the method one rung down a demand ladder (`STRUCTURE_DEMAND`) to the highest lower-demand method allowed that day; applies to the day's default too. Audit `STRUCTURE_READINESS_DOWNGRADE`. Set-reduction still stacks on top.

## Data model

| STORE                                            | HOLDS                                   | ACCESS                                                      |
|--------------------------------------------------|-----------------------------------------|-------------------------------------------------------------|
| <code>teams.strength_send_mode</code>            | team default: individualised · standard | coach                                                       |
| <code>teams.strength_auto_send</code>            | opt-in morning send (bool)              | coach                                                       |
| <code>team_strength_palette.slots</code>         | per-slot exercise pool (jsonb)          | RLS <code>coach_team_ids()</code> / <code>is_staff()</code> |
| <code>team_strength_palette.md_structures</code> | MD → method map (jsonb)                 | same                                                        |
| <code>player_today_strength_override</code>      | the sent session · 1 row/player/date    | coach-written                                               |
| <code>strength_session_overrides</code>          | coach manual swaps                      | coach, team-scoped                                          |
| <code>player_exercise_swaps</code>               | player swaps (log + coach window)       | player-owned · coach read                                   |

## Endpoints

| ROUTE                                             | DOES                                                                |
|---------------------------------------------------|---------------------------------------------------------------------|
| GET/POST /api/coach/team/strength-palette         | palette slots + MD structures (POST merges only the field sent)     |
| GET/POST /api/coach/team/strength-send-mode       | team default mode + auto-send                                       |
| POST /api/coach/team/send-strength-sessions       | bulk send (per-send mode override)                                  |
| POST /api/coach/player/[id]/send-strength-session | single send                                                         |
| POST /api/coach/player/[id]/strength-override     | coach swap — safe alternatives first                                |
| POST/DELETE /api/player/exercise-swap             | player swap — validates target ∈ alternatives, logs, rewrites today |
| GET /api/strength/auto-send                       | Vercel cron 0 6 * * * · opt-in · skip-if-sent · strength days       |

## Audit rules — surfaced as “Why these changes?”

|                        |                      |                                  |
|------------------------|----------------------|----------------------------------|
| STANDARD_MODE          | TEAM_PALETTE_APPLIED | PALETTE_SYMMETRY_UNILATERAL      |
| STRUCTURE_APPLIED      | CORRECTIVE_MERGED    | CORRECTIVE_CORROBORATES_EMPHASIS |
| COACH_OVERRIDE_APPLIED | INJURY_BLOCK         |                                  |

## Evidence base

Rønnestad 2023 microdosing · Bishop 2020 inter-limb asymmetry · Tufano 2017 cluster sets · Cormie 2011 / Liu 2023 contrast & French contrast · Pareja-Blanco 2017 velocity-loss caps · van Dyk 2019 Nordic (–51% hamstring) · Harøy 2019 Copenhagen (–41% groin) · Martin-García 2018 matchday taper · Oranchuk 2023 isometric length/intensity/intent · Krzysztofik 2023 / Jarosz 2025 isometric conditioning → PAPE · Schaefer & Bittmann 2017 pushing vs holding.